

Validate Sigma3

On receipt of **Msg3**, the responder SHALL perform the following:

1. The responder SHALL generate the **S3K** key.
2. The responder SHALL generate the decrypted data:

```
byte TBEData3_A[] = {}
byte TBEData3_Nonce[13] = /* "NCASE_Sigma3N" */
    {0x4e, 0x43, 0x41, 0x53, 0x45, 0x5f, 0x53, 0x69,
     0x67, 0x6d, 0x61, 0x33, 0x4e}

Success, TBEData3 = Crypto_AEAD_DecryptVerify(
    K = S3K,
    C = Msg3.encrypted3,
    A = TBEData3_A,
    N = TBEData3_Nonce
)
```

3. If the value of **Success** is **FALSE**, the responder SHALL send a **status report**: **StatusReport(GeneralCode: FAILURE, ProtocolId: SECURE_CHANNEL, ProtocolCode: INVALID_PARAMETER)** and perform no further processing.
4. The responder SHALL verify that the **NOC** in **TBEData3.initiatorNOC** and the **ICAC** in **TBEData3.initiatorICAC** fulfill the following constraints:
 - a. The Fabric ID SHALL match the Fabric ID matched during processing of the **Destination Identifier** after receiving Sigma1.
 - b. If an **ICAC** is present, and it contains a Fabric ID in its subject, then it SHALL match the FabricID in the NOC leaf certificate.
 - c. The certificate chain SHALL chain back to the Trusted Root CA Certificate **TrustedRCAC** whose public key was matched during processing of the **Destination Identifier** after receiving Sigma1.
 - d. All the elements in the certificate chain SHALL respect the **Matter Certificate DN Encoding Rules**, including range checks for identifiers such as Fabric ID and Node ID.
5. If any of the validations from the previous step fail, the responder SHALL send a **status report**: **StatusReport(GeneralCode: FAILURE, ProtocolId: SECURE_CHANNEL, ProtocolCode: INVALID_PARAMETER)** and perform no further processing.
6. The responder SHALL verify **TBEData3.initiatorNOC** using:
 - a. **Success = Crypto_VerifyChain(certificates = [TBEData3.initiatorNOC, TBEData3.initiatorICAC, TrustedRCAC])**, when **TBEData3.initiatorICAC** is present, or
 - b. **Success = Crypto_VerifyChain(certificates = [TBEData3.initiatorNOC, TrustedRCAC])**, when **TBEData3.initiatorICAC** is not present.
7. If the value of **Success** is **FALSE**, the responder SHALL send a **status report**: **StatusReport(General-**